

Supplementary methods and secondary checks

Single-band delay–preference non-identifiability

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4 October 2026

1 Reproduction and implemented scope

Run `python simulate.py`, `python verify.py`, and `python build_results.py` in the source folder. All numerical observations are synthetic. Complete scripts are supplied as separate files; their source is not duplicated as manuscript pages. Figure files numbered 1–4 below are secondary checks; files 5–7 appear as Figures 2, 3, and 1, respectively, in the main text. Generated `results_macros.tex` is the numerical source used by the manuscript. Detailed per-realization values and intervals are in `results.json`. Exact dependency versions from the execution environment are in `environment.txt`. Settings are not a human sample-size recommendation.

2 Independent marker channels

Let s be externally known stimulus time and let a, b, c be independently measured arrival, ignition-like, and output markers. Their stipulated means are $a = s + \tau$, $b = a + \tau_{\text{pre}}$, $c = b + \tau_{\text{post}}$. The observation vector $(\eta, a - s, b - s, c - s)$ has Jacobian

$$J = \begin{pmatrix} 1 & \Omega & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

in coordinates $(\alpha, \tau, \tau_{\text{pre}}, \tau_{\text{post}})$, with determinant one. In coordinates $(\eta, \tau, \tau_{\text{pre}}, \tau_{\text{post}})$ the first row becomes $(1, 0, 0, 0)$. Independent rank does not validate a physiological marker. Shared marker errors induce correlated interval errors; without a , $b - s$ mixes travel and pre-marker processing. Intervals are not durations of conscious experience.

Synthetic logistic channels have onset means 12, 21, and 26 ms, transition scale 1 ms, and independent Gaussian noise SD 0.025. Onsets are estimated by squared-error search on a 0.25-ms grid over 0–40 ms. Figure S2 is a single consistency/recovery illustration; it is not repeated-sampling coverage or a neural latency estimate.

3 Ordering centroids and finite-frequency failure

Let token times t_k be permuted by σ and nonnegative calibrated weights q_k sum to one. Define

$$\tau_{\text{ord}} = \sum_k q_k (t_{\sigma(k)} - t_k), \quad F_{\sigma}(\omega) = \sum_k q_k e^{-i\omega t_{\sigma(k)}}.$$

Since $F_{\sigma}(0) = 1$, the derivative of $\arg(F_{\sigma}/F_{\text{id}})$ at zero is $-\tau_{\text{ord}}$. Uniform weights give zero for every permutation of the same slots. At finite frequency, wrapping, zeros, and higher moments

invalidate interpretation as a constant delay. Figure S3 uses 12 slots 20 ms apart and weights proportional to 1–12. This mathematical timing statistic is neither a semantic dictionary nor a separately identified conduction latency. It is excluded from the primary confirmatory claim.

4 Serial transfer and kernel directions

For a single acyclic path with no extra inputs and no processing kernels, write $x_k(t) = r_k(t)T_k x_{k-1}(t - \tau_k)$. Let $d_k = \sum_{j=k+1}^m \tau_j$. Repeated substitution gives

$$x_m(t) = \left[\prod_{k=1}^m r_k(t - d_k) \right] (T_m T_{m-1} \cdots T_1) x_0 \left(t - \sum_{k=1}^m \tau_k \right).$$

Matrices preserve path order; gates use retarded times. A feature annihilated along this path remains absent. Additional inputs and bypasses invalidate that isolation assumption. With processing kernels and constant gains, path composition adds convolution of the kernels; time-varying gates generally prevent replacing the process by a single constant latency.

5 Conditional sensitivity and the stipulated learning split

The noise-location curves in Figure S4 implement the main text’s Gaussian class means $\pm rhC/2$. With positive downstream noise, sensitivity rises with gain. Pure upstream noise gives a constant for positive gain and becomes degenerate at exactly zero gain. The derivative of a saturated readout has mixed term

$$\frac{\partial^2 \tanh(wrC)}{\partial r \partial C} = w \operatorname{sech}^2(z) [1 - 2z \tanh z], \quad z = wrC.$$

The sign changes near 0.772, so positive interaction is not an unconditional prediction.

For the learning rule, $u = 1$, target $v = 0.8$, initial trace zero, step size 0.1, and open learning/gain give trace $0.8(1 - 0.9^5)$ after five updates. The closed learning flag leaves the trace exactly zero. Access is $0.8g(\Delta)$, giving 0.8 at zero error and approximately 0.01465 at π . These values are computed from the update rule, not inserted as fitted outcomes. The split is imposed by architecture.

6 Primary simulation settings and uncertainty

The scalar interaction experiment uses 60 realizations of 500 training and 500 test trials, exact observed features, and independent Gaussian noise. Conditional normal-approximation intervals use OLS residual variance with $n - p$ degrees of freedom. Empirical coverage is reported, not asserted equal to nominal coverage; 60 realizations provide a limited Monte Carlo estimate. These intervals do not propagate dictionary, phase, or decoder uncertainty. The quadratic model contains the bilinear model, intentionally demonstrating predictive nonuniqueness.

The temporal experiment has one 900-sample realization with a 30-sample burn-in; 500/150/220 consecutive samples are used for training/validation/test. Gains and content span each segment. Lag selection uses only validation responses. Past inputs needed to predict a test outcome are available, but no test response contributes to fitting. Known maps and kernels are supplied. Temporal excitation is required; a constant signal cannot identify its delay.

The sensor experiment uses 24 realizations, matched observation-noise streams across seven ablations, 80 training and 80 independent test trials, and a fixed six-harmonic Fourier comparator. Gate offsets are searched on 49 points over $[-\pi, \pi]$, concentration on 32 points over $[0.25, 8]$, and nonnegative gain is optimized analytically for each template. Concentration is

grid limited, not an unbiased estimator. Fourier regression has 13 coefficients. The 16-Hz harmonic is exactly rejected by the 6–10 Hz filter for the specified stationary periodic waveforms. The two-cycle alternative generator is an output-transfer function, distinct from adding a 16-Hz recording component. Model wins are predictive comparisons, not calibrated hypothesis-test decisions.

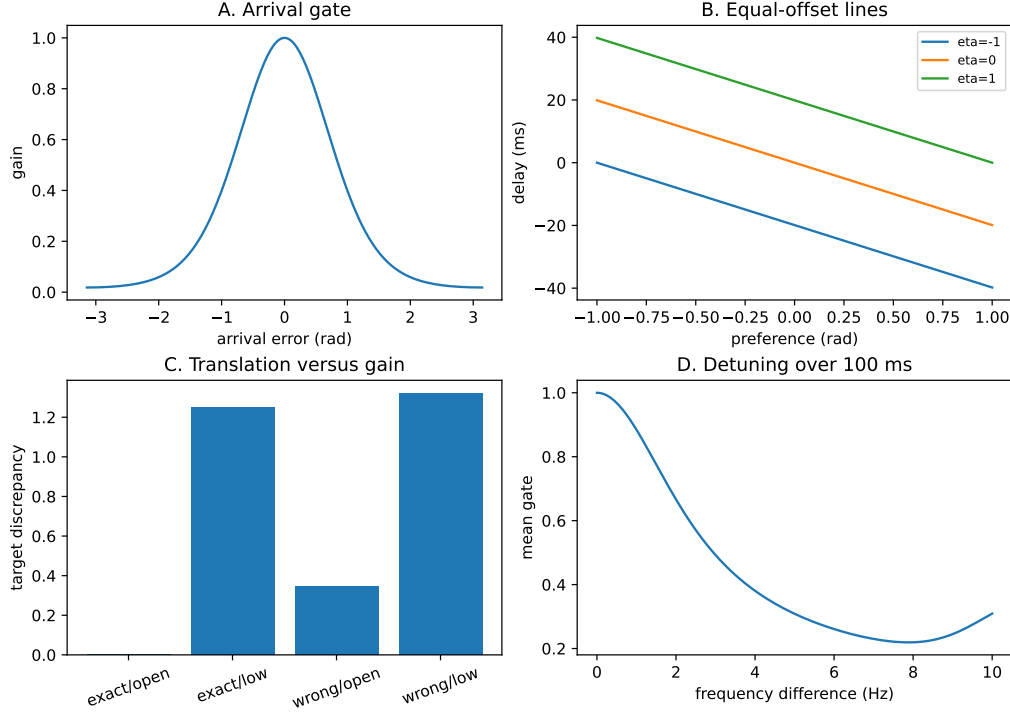


Figure S1: Secondary consistency checks: phase gate, parameter ridge, translation error versus gain, and detuning. Negative delays in the ridge display are an algebraic extension; only non-negative admissible delays enter the physiological model.

7 Main-text simulation plots

8 Reproduction of the observation model

Run `python reproduce.py` in the source folder. It checks four identities and writes `reproduction_checks.json`.

The transferred cross-spectrum has phase $-(\alpha + \Omega\tau)$. Shifts that preserve this sum leave the phase unchanged to numerical precision. The simultaneous oscillator contrast is a constant in that script and does not contain α or τ . That is why a phase-locking score on the raw contrast does not establish the title claim.

On the rank-one map, kernel amplitude is identically zero and the image crosses 0.25 exactly when the gate does. A moving offset of slope -1.6 rad/s first clears 0.5 rad at 0.562 s. A 40 ms pulse puts the mixed-term peak at 50 ms, and a closed unreported gate returns that amplitude to zero.

The exact-feature coupling intervals covered in 52 of 60 runs (86.7 percent). The nominal 95 percent level is not a claim that the interval formula is mis-specified. Feature noise of SD 0.15 drops coverage to 0 percent. Both numbers are regenerated by `simulate.py` and checked by `verify.py`. No public deposit DOI is claimed; the scripts are in this archive.

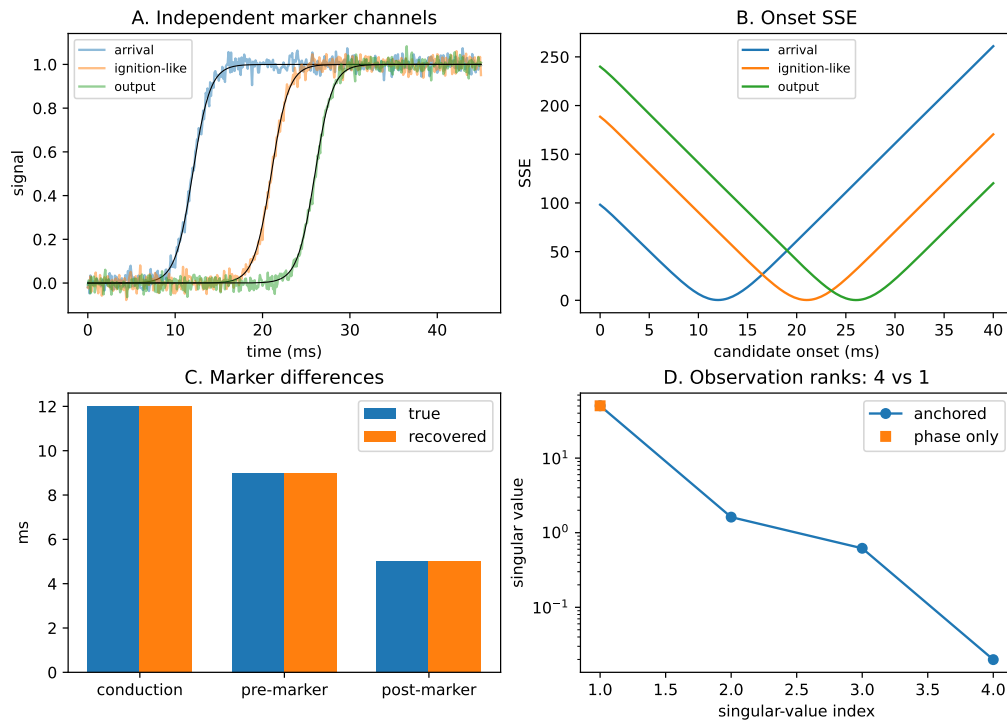


Figure S2: Stipulated independent logistic marker channels and their observation rank. Singular values use the displayed units; their magnitudes are not unit-invariant conditioning measures.

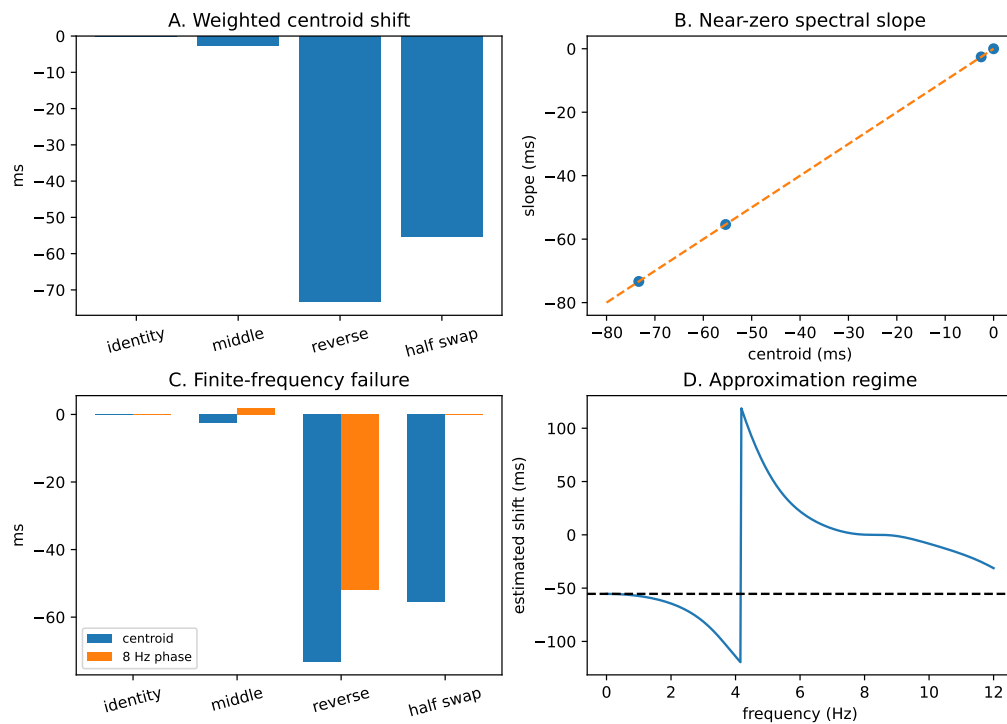


Figure S3: Ordering centroid equals the near-zero spectral slope; a finite-frequency phase-derived shift can disagree.

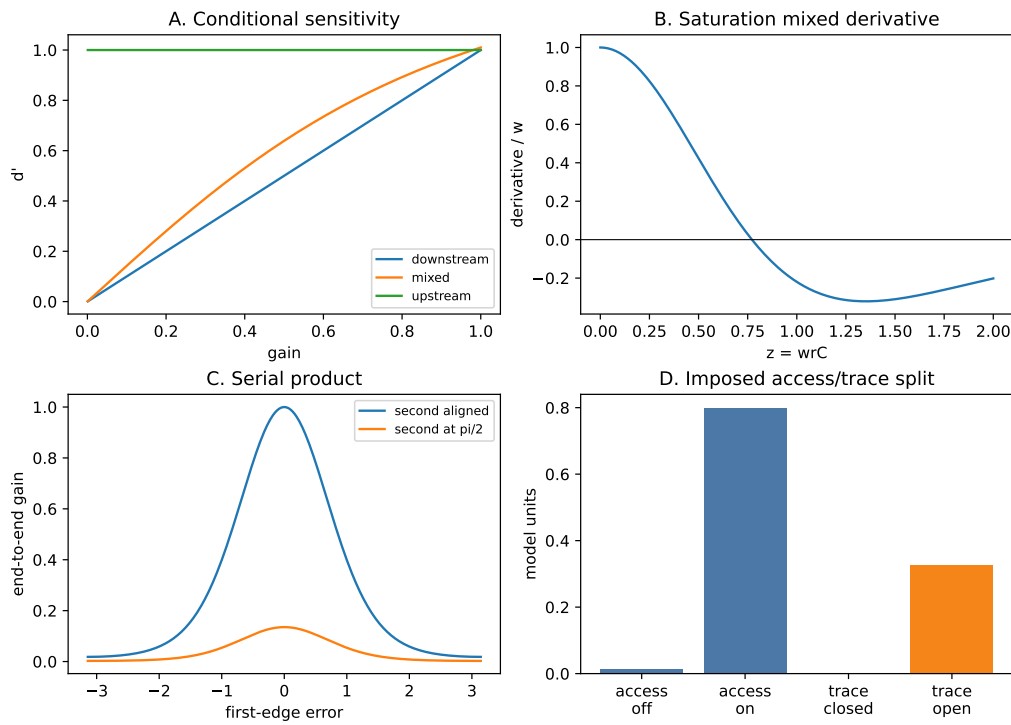


Figure S4: Conditional sensitivity, saturation, retarded serial gain, and the architecture-imposed access-trace split.

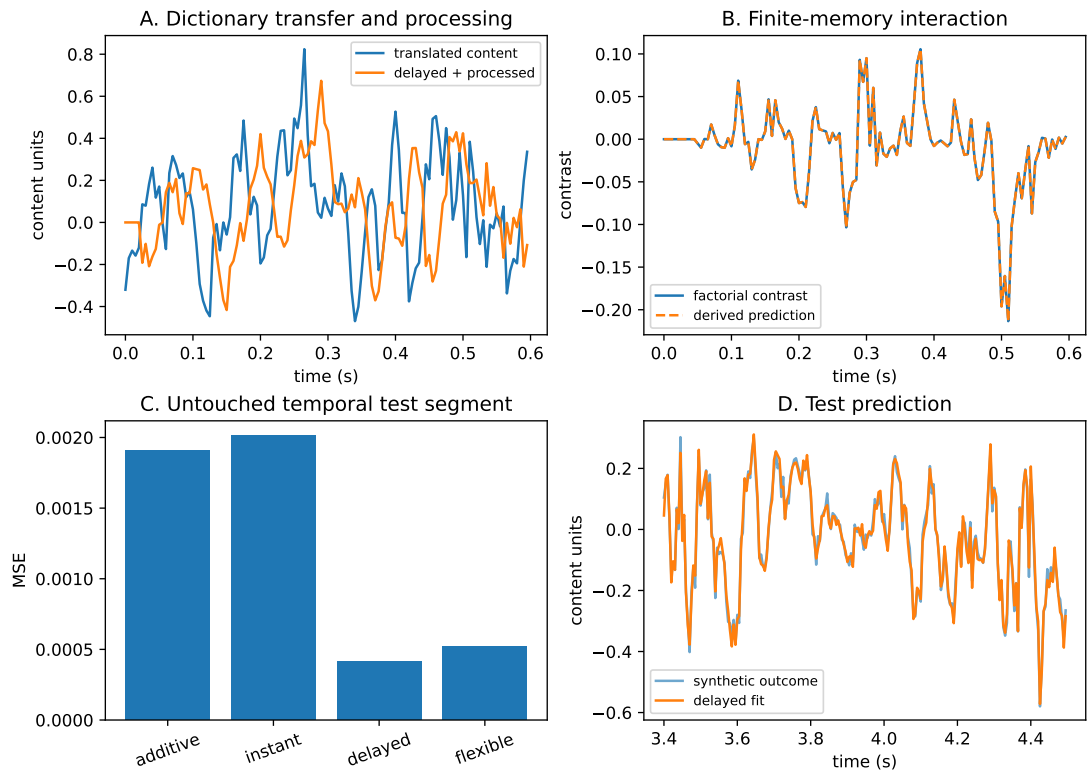


Figure S5: Delayed dictionary translation, factorial contrast, held-out prediction, and test fit.

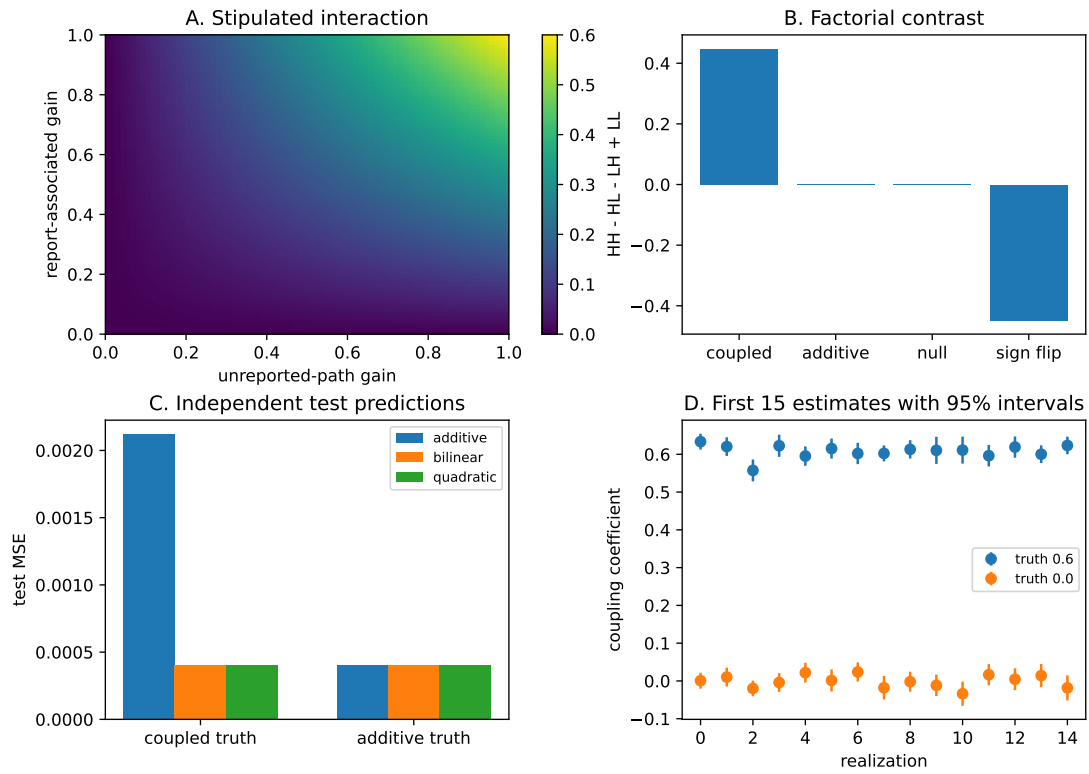


Figure S6: Restricted interaction recovery. Panel D shows the first 15 of 60 realizations.

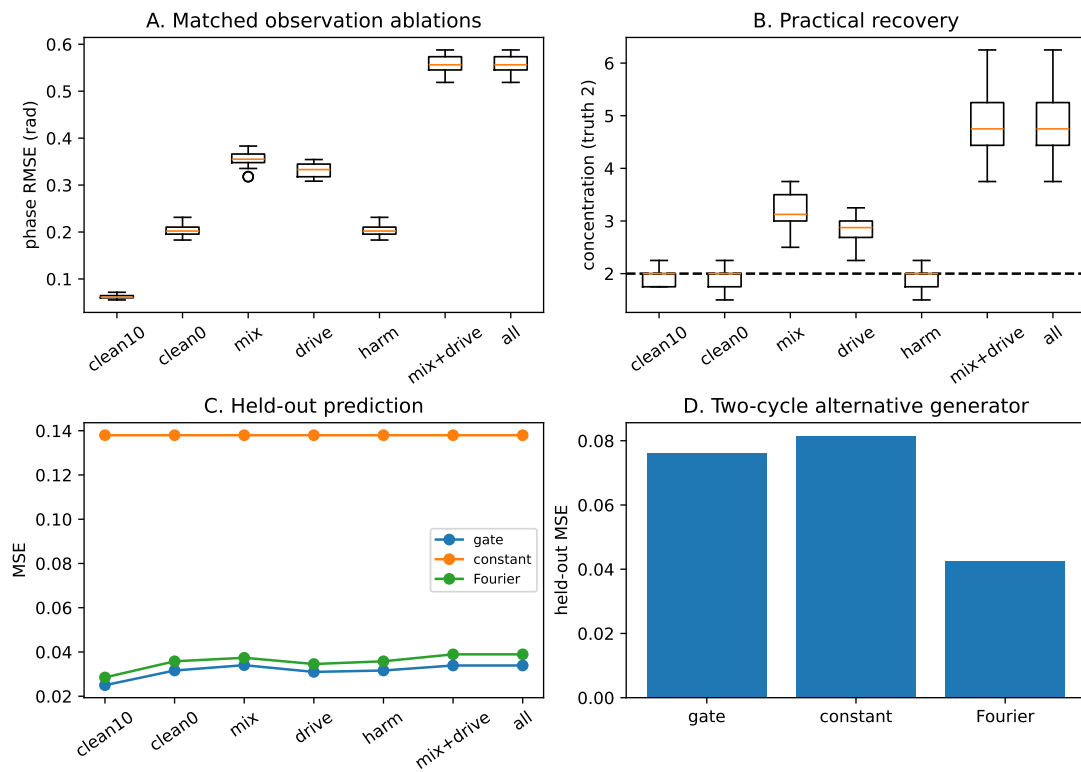


Figure S7: Matched observation ablations and the two-cycle alternative generator.

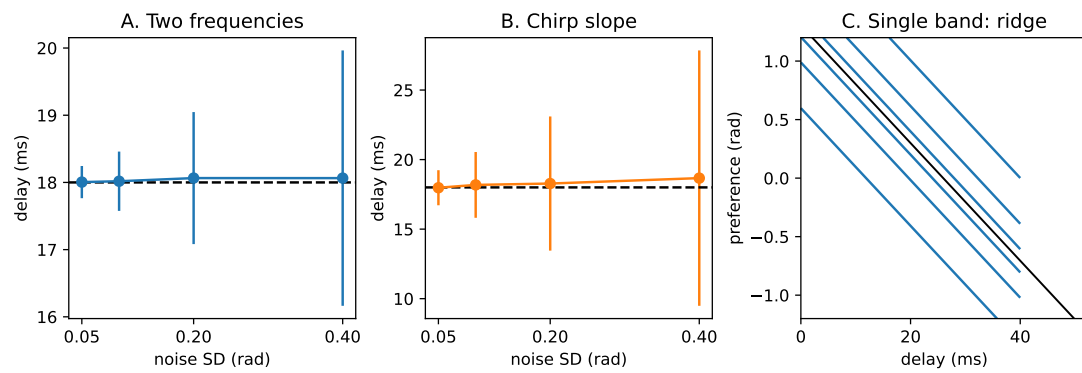


Figure S8: Constructive escapes. Two frequencies and a chirp recover delay. The single-band panel is constant along the ridge.